

SYNOPSIS REPORT

On

Database Management System Workbench

Submitted in partial fulfillment of the requirements of the degree of

BACHELOR OF BUSINESS ADMINISTRATION (COMPUTER APPLICATION)



Submitted by

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CERTIFICATE

This is to certify that the Project Report entitled **Database Management System Workbench** prepared by **Suraj Uttamchand Oswal** students of Bachelor of Business Administration (Computer Application) semester-6 course at Arihant College of Art's, Commerce and Science(ACACS), Pune-01.

To the best of our knowledge, this is original study done by the said student and important sources used by them have been duly acknowledged in this report.

The report is submitted in partial fulfillment BBA-CA Semester-6 syllabus as per rule prescribed guideline of Savitribai Phule Pune University.

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❖ Name of Project:

Database Management System Workbench

❖ Introduction

In today's digital world, databases play an important role in storing and managing data efficiently. Almost every application, whether it is a website, mobile app, or business software, depends on a database to organize and retrieve information. While there are many professional tools available to manage databases, such as MySQL Workbench and phpMyAdmin, they can sometimes feel complex for beginners and students who are still learning database concepts.

As a student of Computer Applications, I wanted to build something that not only helps in understanding DBMS concepts but also gives practical experience in full-stack development. Therefore, I developed a Web-Based DBMS Workbench that allows users to create databases, create tables with constraints, insert, update, and delete records, and execute SQL queries through a graphical interface.

The main purpose of this project is to combine theoretical knowledge of Database Management Systems with real-world implementation. Through this project, I have applied concepts such as primary keys, constraints, CRUD operations, and SQL query execution in a practical environment. The system is developed using HTML, CSS, JavaScript, AngularJS for the frontend, and Node.js with MySQL for the backend.

This project helped me understand how frontend and backend communicate with each other, how database queries are executed securely, and how a user-friendly interface can simplify complex database operations. It also improved my understanding of system design, validation rules, and real-time data handling.

Overall, this project represents my effort to create a simplified, interactive, and educational database management tool that demonstrates both DBMS concepts and web development skills.

❖ Existing System

In the existing system, database management is usually done using professional tools such as MySQL Workbench, phpMyAdmin, or by using command-line interfaces. These tools are very powerful and provide many advanced features for managing databases, tables, relationships, and queries.

However, for beginners and students, these tools can sometimes be difficult to understand because they contain many complex options and configurations. A new learner may feel confused while performing basic operations like creating tables with constraints or understanding how primary keys and foreign keys work. Also, command-line-based database management requires strong knowledge of SQL syntax, and even a small mistake in query writing can cause errors.

Another limitation of the existing system is that most tools focus on professional-level usage rather than educational purposes. They do not always provide a simplified and interactive environment specifically designed for learning database concepts step by step. In addition, some desktop-based tools require installation, configuration, and proper system setup, which may not be convenient for every user.

Therefore, although existing systems are powerful and widely used in industries, they may not be ideal for students who are still learning DBMS concepts and want a simpler, more interactive interface to understand how databases work internally.

❖ Proposed System

To overcome the limitations of the existing system, I have proposed and developed a **Web-Based DBMS Workbench** that provides a simple, interactive, and user-friendly interface for managing databases. The main idea of the proposed system is to make database operations easier to understand and perform, especially for students and beginners.

In this system, users can create and delete databases through a graphical interface without writing complex SQL commands manually. It also allows users to create tables by selecting data types and applying constraints such as Primary Key, Not Null, Unique, Unsigned, and Auto Increment using simple form options. This makes it easier to understand how table structures are designed.

The proposed system also supports full CRUD operations (Create, Read, Update, Delete). Users can insert new records, update existing records by double-clicking them, and delete records with confirmation. This provides a practical understanding of how data manipulation works in real databases.

Additionally, the system includes a built-in SQL query editor where users can execute custom SQL queries. It also provides a structured SELECT query builder that helps users generate queries using input fields instead of writing the entire query manually. This feature helps beginners learn SQL syntax in a guided way.

Another important feature of the proposed system is exporting table data to PDF format, which makes it useful for reporting purposes. The system also includes search functionality, keyboard shortcuts, dark/light theme support, and draggable pop-up windows to improve user experience.

The proposed system is developed using HTML, CSS, JavaScript, and AngularJS for the frontend, and Node.js with MySQL for the backend. It follows a client-server architecture where the frontend communicates with the backend through REST APIs to perform database operations securely.

Overall, the proposed system provides a simplified and educational database management tool that combines theoretical DBMS concepts with practical implementation in a web-based environment.

❖ Scope

- To create and delete databases using a graphical interface.
- To create and delete tables with different data types and constraints.
- To perform CRUD operations (Insert, Update, Delete, View data).
- To execute SQL queries through a built-in SQL editor.
- To provide a structured SELECT query builder for easier query formation.
- To display table metadata such as Primary Key (PK), Foreign Key (FK), and Not Null (NN).
- To export table data in PDF format.
- To implement keyboard shortcuts and theme switching (Dark/Light mode).
- To demonstrate client-server communication using Node.js and MySQL.

❖ Objectives

- To understand and apply DBMS concepts in a practical way.
- To implement database creation, table design, and data manipulation features.
- To gain hands-on experience in full-stack development (Frontend + Backend).
- To understand how SQL queries are executed from a web application.
- To learn how to connect a Node.js backend with a MySQL database.
- To design a user-friendly interface for database operations.
- To improve problem-solving and logical thinking skills.
- To develop a mini-DBMS Workbench similar to tools like MySQL Workbench for learning purposes.

❖ Modules

1. Database Management Module

- Create and delete databases.
- Display and search available databases.
- Validate database names.

2. Table Management Module

- Create tables with different data types.
- Apply constraints like Primary Key, Not Null, Unique, etc.
- Add or remove columns dynamically.
- Delete tables.

3. Data Management (CRUD) Module

- Insert new records.
- View table data.
- Update records.
- Delete records using Primary Key.

4. SQL Query Module

- Execute custom SQL queries.
- Provide a structured SELECT query builder.
- Display results in the output window.

5. PDF Export Module

- Export table data to PDF format.

6. User Interface Module

- Dark/Light theme toggle.
- Search functionality.
- Keyboard shortcuts.
- Draggable pop-up windows.

❖ Software Requirements

- **Operating System:** Windows 10 or above (can also run on Linux or macOS)
- **Frontend Technologies:** HTML, CSS, JavaScript, AngularJS
- **Backend Technology:** Node.js
- **Server Framework:** Express.js
- **Database:** MySQL
- **Database Driver:** MySQL2 (Node.js package)
- **PDF Generation Library:** PDFKit
- **Browser:** Google Chrome / Microsoft Edge / Any modern web browser
- **Code Editor:** Visual Studio Code

❖ Hardware Requirements

- **Processor:** Intel i3 or above
- **RAM:** Minimum 4 GB (8 GB recommended)
- **Storage:** At least 1 GB free disk space
- **System Type:** 64-bit system recommended
- **Internet Connection:** Required for installing packages and libraries

❖ Application

- Can be used by students to understand DBMS concepts practically.
- Helpful for practicing SQL queries in a simple and interactive way.
- Useful for small academic projects that require database management.
- Can be used in computer labs for learning database creation and table design.
- Suitable for beginners who want to learn database operations without using complex tools.
- Can be used to demonstrate CRUD operations in web development projects.
- Useful for generating simple reports by exporting table data to PDF.
- Can act as a mini alternative to tools like MySQL Workbench for educational purposes.

❖ Merits

- Provides a simple and user-friendly interface for database management.
- Helps students understand DBMS concepts practically.
- Supports full CRUD operations (Insert, Update, Delete, View).
- Allows execution of custom SQL queries.
- Includes structured SELECT query builder for beginners.
- Displays Primary Key (PK), Foreign Key (FK), and Not Null (NN) information clearly.
- Provides PDF export features for table data.
- Includes dark and light theme for better user experience.
- Reduces dependency on complex professional tools like MySQL Workbench.
- Lightweight and runs in a web browser

❖ Feasibility Study

1. Technical Feasibility

- The project uses commonly available technologies like HTML, CSS, JavaScript, Node.js, and MySQL.
- All required software tools are open-sourced and easily accessible.
- The system can run on a normal computer with basic configuration.
- The development environment is simple and does not require expensive setup.
- Therefore, the project is technically feasible.

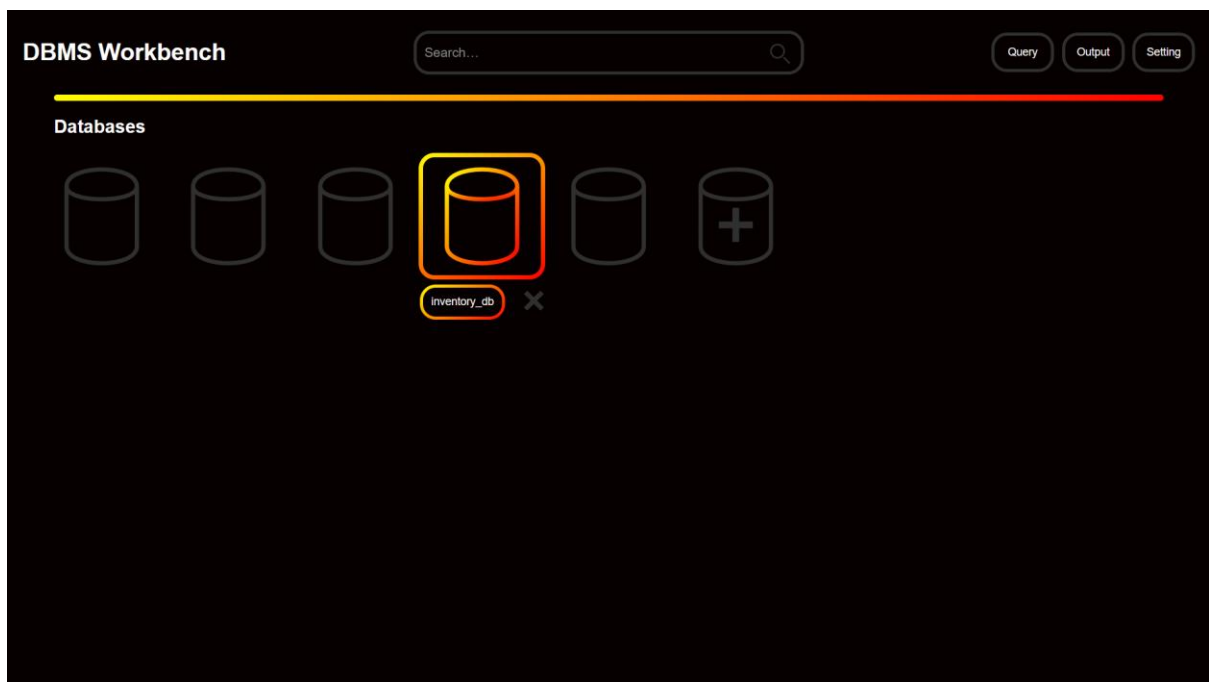
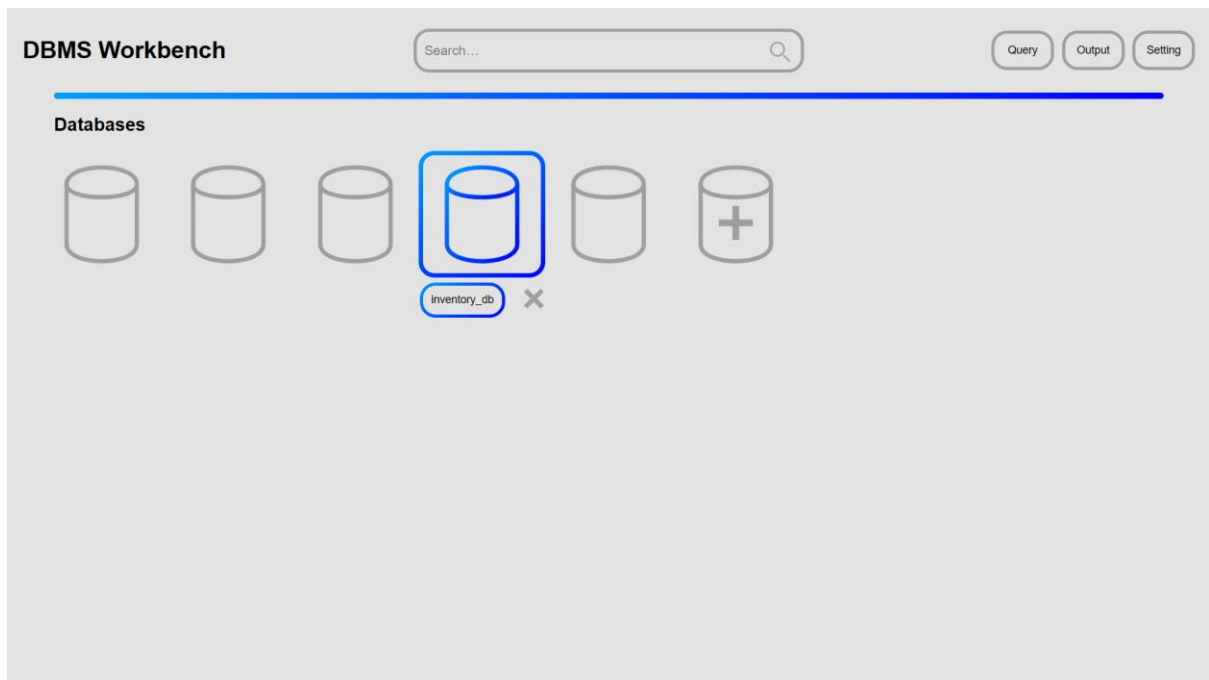
2. Economic Feasibility

- The project does not require any paid software.
- Node.js, MySQL, and other tools are free to use.
- No additional hardware investment is required.
- The development cost is very low.
- Hence, the project is economically feasible.

3. Operational Feasibility

- The system is user-friendly and easy to understand.
- It provides a simple graphical interface for performing database operations.
- Users do not need advanced technical knowledge to operate the system.
- The system helps students learn database management practically.
- Therefore, the project is operationally feasible.

❖ Output Screenshots



DBMS Workbench

Search...

Query

Output

Setting

World database tables



DBMS Workbench

Search

Query

Output

Setting

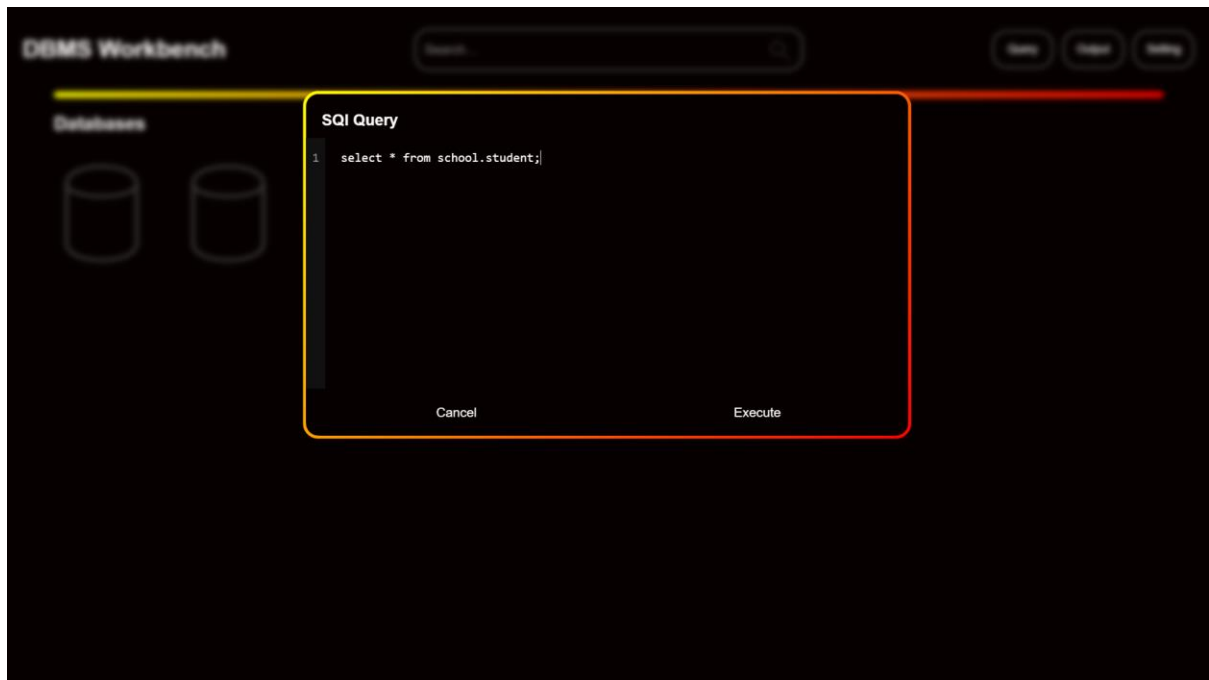
Data from city table in world database

Add Row

Update Data

Print Table

ID (PK) (NN)	Name (NN)	CountryCode (FK) (NN)	District (NN)	Population (NN)
 1	Kabul	AFG	Kabul	1780000
 2	Qandahar	AFG	Qandahar	237500
 3	Herat	AFG	Herat	186800
 4	Mazar-e-Sharif	AFG	Balkh	127800
 5	Amsterdam	NLD	Noord-Holland	731200
 6	Rotterdam	NLD	Zuid-Holland	593321
 7	Haag	NLD	Zuid-Holland	440900
 8	Utrecht	NLD	Utrecht	234323
 9	Eindhoven	NLD	Noord-Brabant	201843
 10	Tilburg	NLD	Noord-Brabant	193238



Create New Table

Table Name

student

id	INT()	size	Primary Key	Not Null	Unique	Unsigned	Auto Increment	disabled
name	VARCHAR()	50	Primary Key	Not Null	Unique	Unsigned	Auto Increment	Enter Default Value / Expression
address	VARCHAR()	150	Primary Key	Not Null	Unique	Unsigned	Auto Increment	pune
phone	INT()	size	Primary Key	Not Null	Unique	Unsigned	Auto Increment	Enter Default Value / Expression
gender	CHAR()	size	Primary Key	Not Null	Unique	Unsigned	Auto Increment	Enter Default Value / Expression
+			-					

Reset

Create

DBMS Workbench

Run

Search

Query

Output

Setting

Data from city table in world database

Add Row

Update Data

Print Table

ID (PK) (NN)	Name (NN)	CountryCode (FK) (NN)	District (NN)	Population (NN)
1024	Mumbai (Bombay)	IND	Maharashtra	10500000
1025	Delhi	IND	Delhi	7206704
1026	Calcutta [Kolkata]	IND	West Bengali	4399819
1027	Chennai (Madras)	IND	Tamil Nadu	3841396
1028	Hyderabad	IND	Andhra Pradesh	2964638
1029	Ahmedabad	IND	Gujarat	2876710
1030	Bangalore	IND	Karnataka	2660088
1031	Kanpur	IND	Uttar Pradesh	1874409
1032	Nagpur	IND	Maharashtra	1624752
1033	Lucknow	IND	Uttar Pradesh	1616115

Select

all

Form

city

Where

CountryCode = "IND"

Group By

enter column name

Having

write condition

Order By

column ASC | DESC

Limit

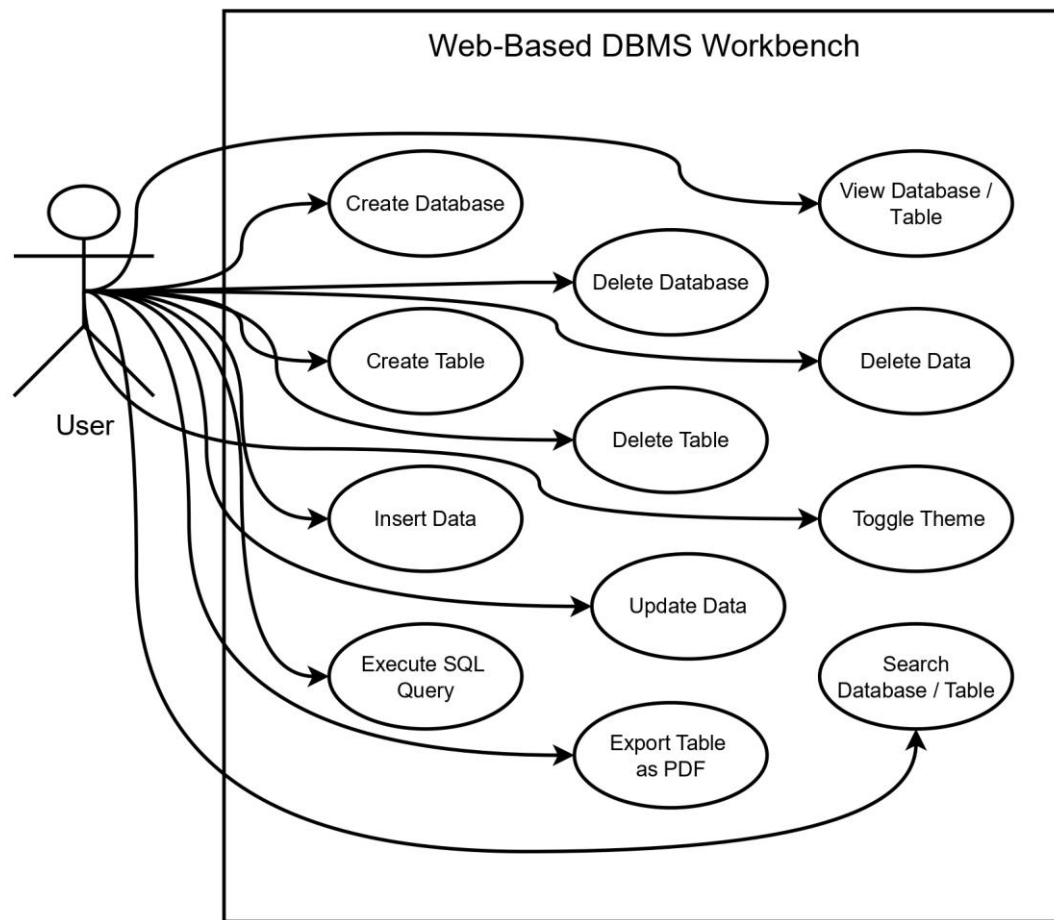
enter limit number

❖ UML Diagram

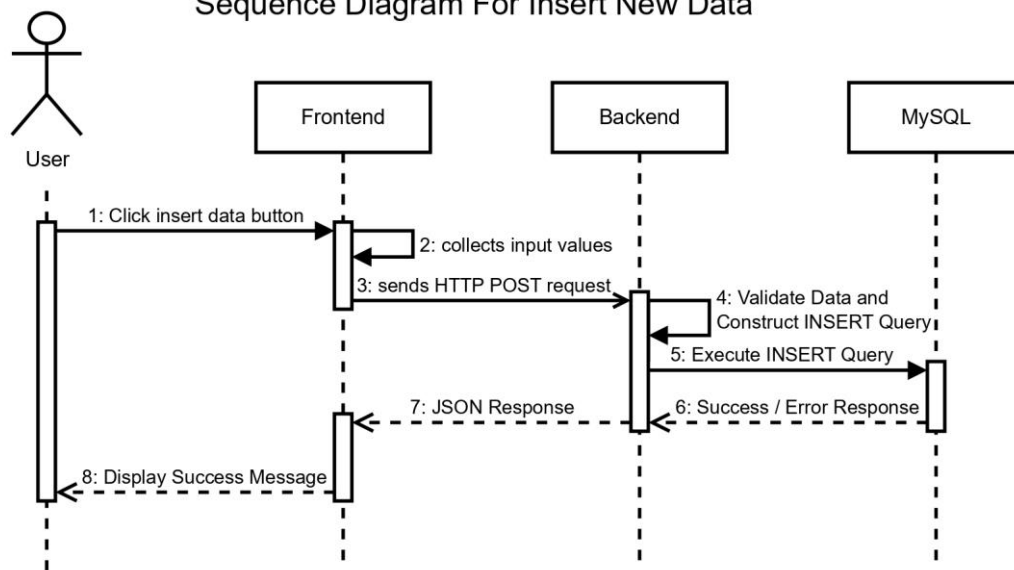
Class diagram of DBMS Workbench



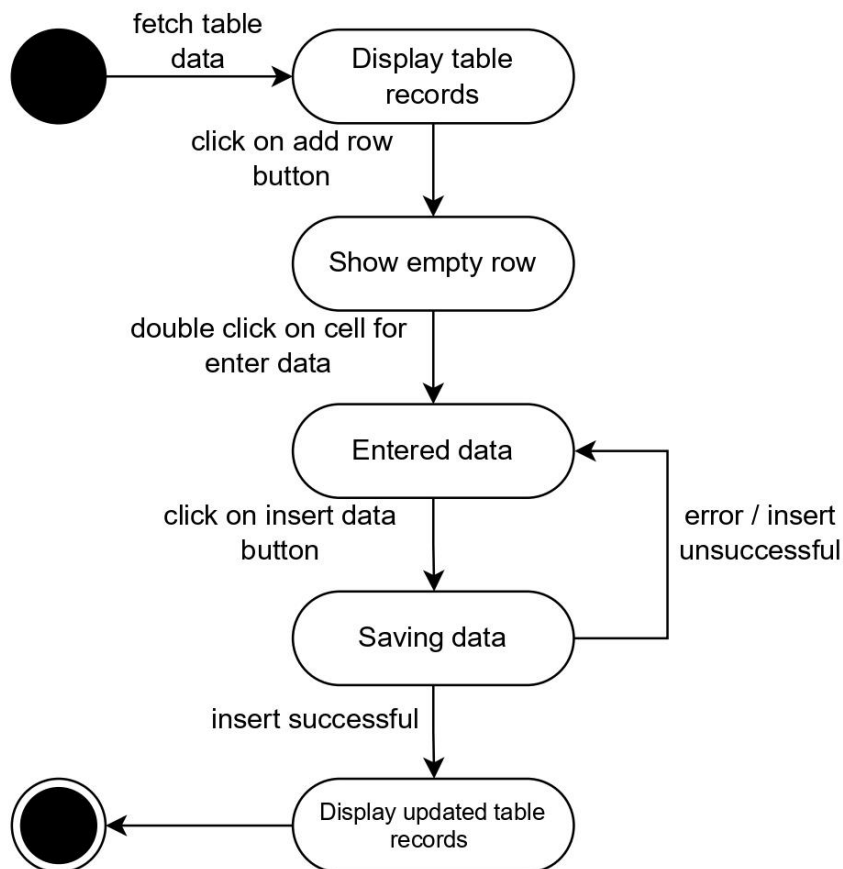
Use Case Diagram Of DBMS Workbench



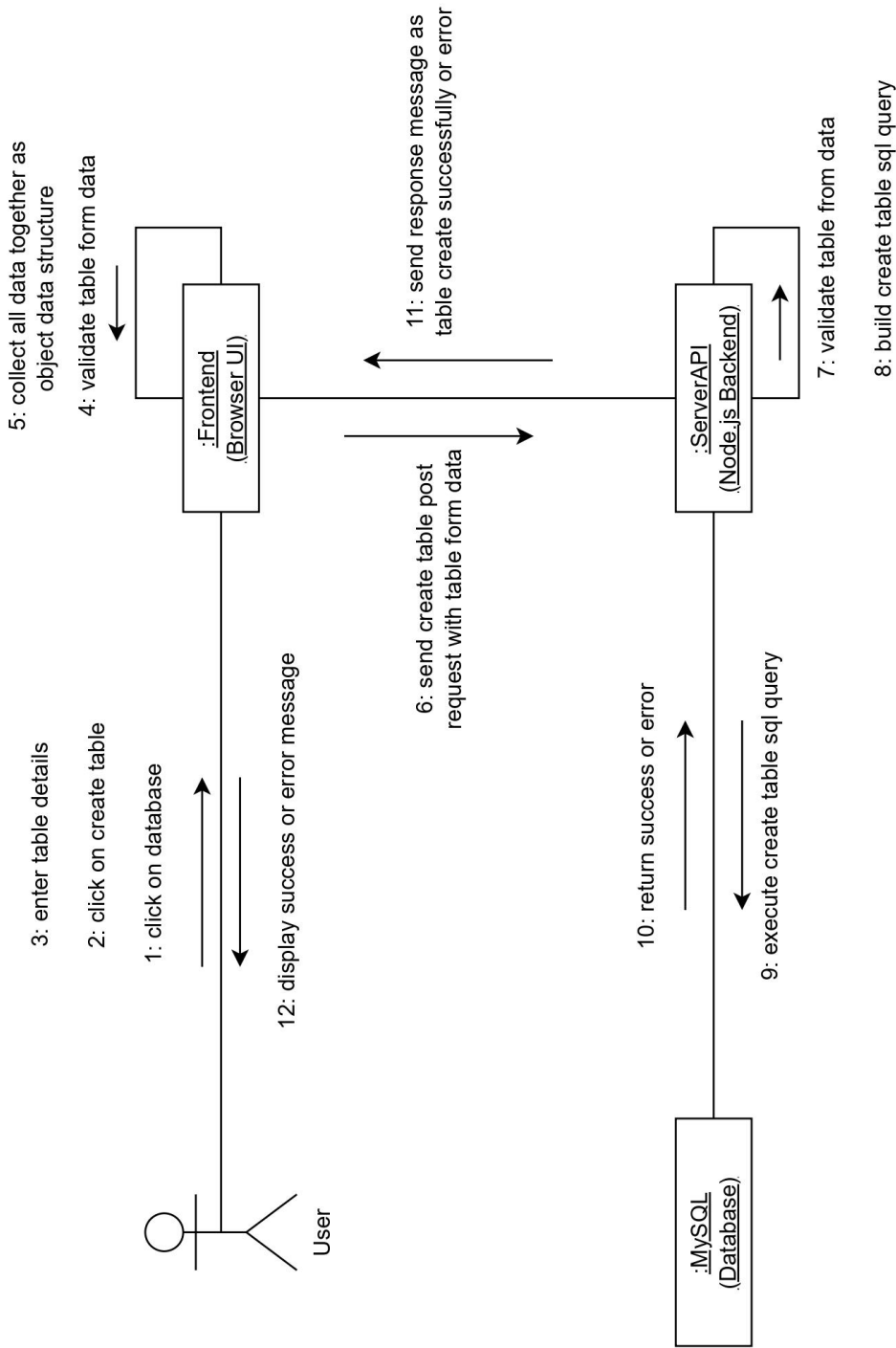
Sequence Diagram For Insert New Data



State Chart Diagram Of Insert Table Data

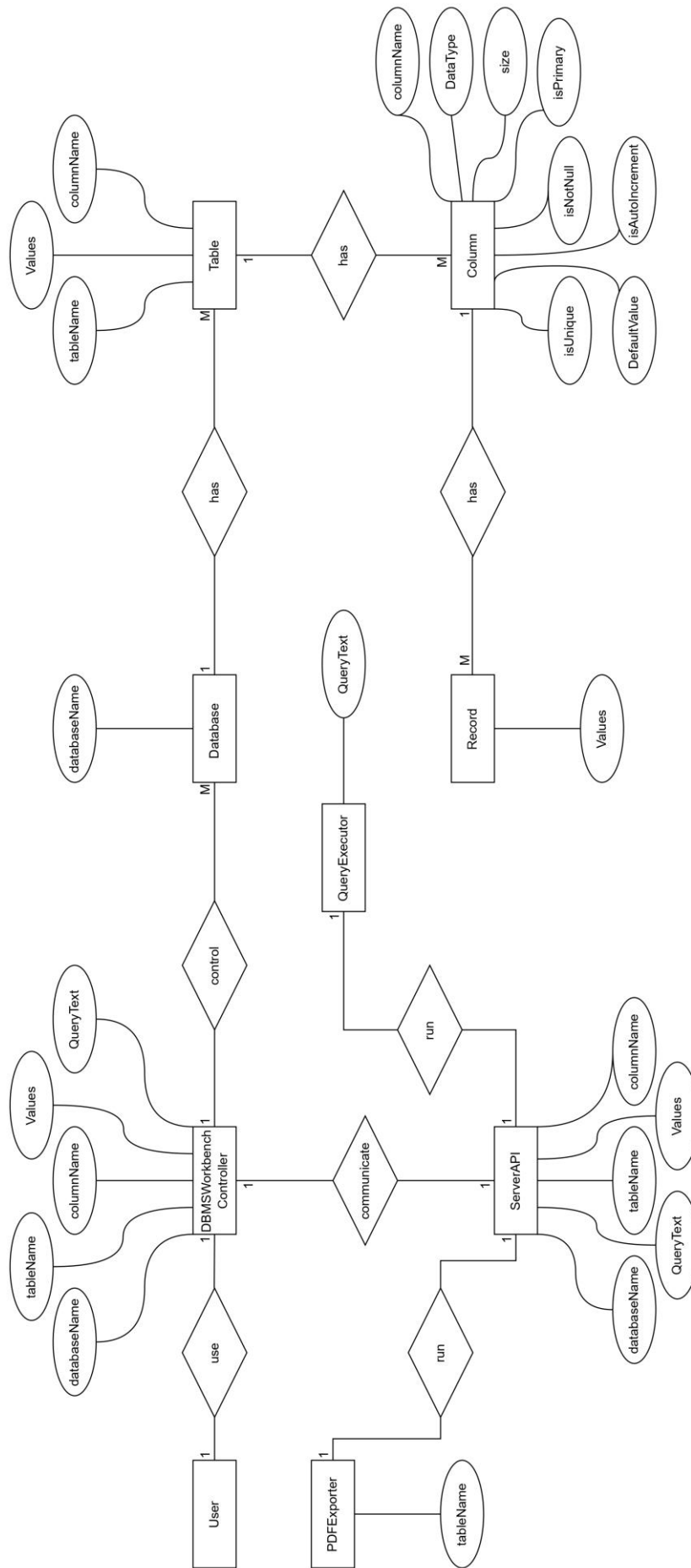


Collaboration Diagram Of Table Creation



❖ E-R diagram

ER Diagram Of DBMS Workbench



❖ **Limitations / Demerits**

- Does not include user authentication or login system.
- No role-based access control (all users have the same permissions).
- Does not support advanced database features like stored procedures or triggers.
Cannot manage multiple database servers at once.
- No graphical ER diagram generation.
- Limited to the local MySQL server setup.
- Not suitable for large-scale enterprise use

❖ **Future Enhancements**

- Add user authentication and login to secure access.
- Implement of role-based access control (Admin, User, etc.).
- Add support for stored procedures and triggers.
- Provide graphical ER diagram generation features.
- Add table structure editing (modify existing columns).
- Implement database backup and restore functionality.
- Support connections to remote or multiple database servers.
- Add data import and export options (CSV, Excel, etc.).
- Improve UI design to make it more professional and responsive

❖ Conclusion

In this project, I have successfully developed a Web-Based DBMS Workbench that allows users to manage databases, tables, and records through a simple graphical interface. The main objective of this project was to understand and apply Database Management System concepts in a practical way, and this goal has been achieved.

Through this project, I implemented important DBMS concepts such as database creation, table structure design, primary keys, foreign keys, constraints, and CRUD operations. I also developed features like SQL query execution, structured SELECT query builder, and PDF export functionality. This project helped me understand how frontend and backend communicate with each other and how database queries are processed securely.

While developing this system, I gained practical knowledge of full-stack development using HTML, CSS, JavaScript, AngularJS, Node.js, and MySQL. I also improved my problem-solving skills and learned how to handle validation, routing, and dynamic data rendering.

Although the system has some limitations, it works efficiently for learning and small-scale database management purposes. Overall, this project has enhanced my understanding of DBMS concepts and web application development, and it has given me confidence in building real-world applications.

❖ References / Bibliography

AI: ChatGPT

YouTube

MDN Web Docs